2025 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM) | 979-8-3315-2521-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/IEEM63636.2025.11357588

Robust Unsupervised Speaker-profile Analytics with Large Language Model-embedding-based Text Mining and Outlier Detection for Seed-stage High-tech Startup Recruiting

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Abstract – **Efficient identification of individuals who can become core members of an organization while adapting to uncertain organizational changes is critically important in the seed stage in startups, where it is often difficult to allocate sufficient time for hiring and training. Although previous studies have investigated such identification based on utterances during meetings, these approaches lack robustness against variations in meeting participants and contexts, making their practical implementation highly challenging. Consequently, this study proposes an analysis method based on clustering with large language model-derived embedding vectors, which remain robust even when speakers or meeting topics change. Furthermore, the trained weights can be easily reused, enabling organizations to perform such analyses at low cost and with minimal effort.**

Keywords - **High-tech startups, human resource management, large language model, text mining**

I. INTRODUCTION

The seed stage in high-tech startups is generally accepted as the riskiest and most fluid phase in the startup lifecycle, with organizational size often undergoing significant changes [1][2][3][4]. Consequently, such organizations should hire personnel who are well-suited to the current stage and growth trajectory, and minimize risks by proactively terminating employment of unsuitable hires. Previous studies [5] proposed capability assessment frameworks designed to enable the identification of core members who can withstand large organizational changes and external environmental uncertainty.

However, although the use of principal component analysis (PCA) in previous studies[5] provided satisfactory interpretability, it was limited in that the projection space itself could become unsuitable if the participants' speech patterns changed significantly, making it difficult to use the same set of weights consistently. The type of meeting—whether a development progress meeting, hiring interview, or one-on-one session—and composition of participants—such as executives and engineers, engineers and managers, or executives and managers—can significantly affect the level of abstraction in the language used and amount of contextual background. Therefore, when using PCA a set of reference weights specific to the organization should be established initially, and updated as the organization grows or changes personnel. This implies practically that the approach lacks robustness, even within the same organization.

Furthermore, as indicated in [5][6], if the weights are updated in an environment exclusively comprising engineers with a low tolerance for uncertainty—who engage solely in discussions about technical terms and fail to consider upstream business perspectives, customer needs, or market trends—no immediate problem may emerge. However, if the organization predominantly comprises engineers with a high tolerance for uncertainty—who can become core members—the process of updating weights may lead to identifying outliers within this already highly capable group, making it difficult to measure the true tolerance for uncertainty as originally intended. Consequently, stable operation would require relearning the PCA weights for each change in the personnel or meeting context. This highlights the need for a framework that can be used consistently in practical settings with minimal overhead.

II. PURPOSE

In previous studies[5][6], analytical methods were sensitive to fluctuations in the number of participants and conversation content. The PCA weights need to be updated each time to maintain accurate evaluation when personnel are added; moreover, the directionality of these weights may vary during each training phase.

Therefore, this study proposes a framework that is robust to changes in the set of participants under analysis and that can be deployed practically without the need for frequent retuning or retraining tailored to individual participants.

In addition to adopting an approach different from that used in the previous studies, an important objective of this research is to examine the relationship between tolerance for uncertainty and utterance characteristics from a novel perspective.

III. METHODOLOGY

A. Construction of the Word Embedding Space

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A detailed examination of the technical limitations of previous studies[5] revealed that the root cause is in the projection of utterances into a different PCA space for each meeting under analysis, resulting in utterances and words being mapped to different spaces each time. Furthermore, in several previous approaches, words themselves were not vectorized; instead, word frequencies and the relationships between words for everyone, were expressed as one-dimensional vectors, and these frequencies were then analyzed. This approach is sensitive to fluctuations in word frequency and the presence of unusual words, which can significantly affect the distances between vectors based on a single outlier word.

In practice, the meaning of a word can differ across organizations, and the vocabulary used is expected to vary substantially depending on the meeting or organization. Frequency-based text-mining techniques and tools such as KHCoder [7] are highly susceptible to minor changes in context and meeting participants, which can introduce considerable instability into the results. Although forcibly projecting identical words onto the same semantic space across organizations is optional, constructing a relatively stable word embedding space for each organization is desirable.

In this study, we addressed these challenges using the embedding spaces generated by large language models (LLMs). Although this method increases the dimensionality and information content of word representations, it is more robust to contextual changes and the appearance of unusual words. This enables a more flexible and stable word vectorization. For commercial solutions, application programming interfaces (APIs), such as those provided by OpenAI's GPT-4o model can be used to obtain embedding vectors, whereas models such as Sentence-BERT [8] are available as open-source alternatives. In this study, we used the embedding vectors obtained from Sentence-BERT.

B. Dimensionality Reduction Methods for Robust Basis Vector Acquisition

In previous studies[5][6], the basis vectors were typically determined using singular value decomposition (SVD) or PCA for dimensionality reduction and matrix selection. However, the meanings of these basis vectors can be significantly affected by the presence of outliers or unusual words, which is similar to the problems observed in word vectorization. Furthermore, methods such as PCA or SVD, which construct bases by simply enforcing independence among vectors, may be considered too coarse when handling higher-order information, such as LLM-based embedding vectors. It may then be preferable to use more sophisticated techniques that are tolerant of variation and better suited for high-dimensional vectors.

Current approaches apply PCA-based dimensionality reduction and outlier detection after text mining, which makes it challenging to adequately capture changes in the vocabulary itself as well as contextual shifts in the meaning of the same word. Consequently, it is difficult to directly reuse the basis vectors or weights learned from one meeting context in another. This limitation arises because PCA is fundamentally a technique that maximizes the variance in the analyzed dataset, and is therefore highly susceptible to the influence of dominant topics or changes in specific words. Moreover, because PCA only models linear relationships, it does not capture the complex meanings present in words and sentences sufficiently.

This study therefore proposes the use of UMAP [9] as a nonlinear alternative. The proposed framework enables a more stable analysis than those that rely solely on PCA and require frequent updates to weights or basis vectors. In addition, with respect to distance metrics, we propose a novel approach in which the weights remain adaptable, allowing for flexibility in response to significant shifts in meeting topics, while also introducing custom metrics that prevent the misclassification of speakers whose conversational volume is too low. This enables a fair and robust analysis of speaker characteristics.

IV. ANALYSIS

A. Data Preparation

Adansons Inc., where the author serves as the co-founder, provided meeting data for the analysis. Each meeting involved 4–5 participants comprising two executives and two or three engineers. One of the engineers included in the analysis has since left the company owing to differences in values and cultural fit.
All meetings were held using an online conferencing tool, and each session was recorded on video and audio.

The purpose of these meetings was for the two executives, who were knowledgeable about the patented technologies, to instruct other members on their use, discuss specific use cases and implementation strategies, and deliberate on effectively leveraging these technologies in the company's products. These meetings aimed to educate employees on the critical thinking skills required for utilizing the patented technologies as well as the analytical and evaluation skills necessary for handling data.
Each meeting lasted approximately 50 minutes when an average of 10,000 words were spoken.

B. Detailed Analysis Procedure

In this study, we propose a procedure that automatically determines whether speakers have significantly different speaking patterns by simply uploading the recorded meeting data. Furthermore, we introduce a mechanism to prevent speakers with extremely few utterances from being incorrectly identified as having divergent speaking tendencies.

First, as in previous studies, speech transcription was performed using OpenAI's Whisper [10]. For speaker

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diarization, we adopted an open-source pipeline provided by pyannote.audio [11][12]. This pipeline comprises voice activity detection, speaker embedding extraction (based on ECAPA-TDNN), agglomerative hierarchical clustering for speaker clustering, and overlapping speech detection, enabling highly accurate results even in semi-supervised settings. In this study, we used the "pyannote/speaker-diarization" pipeline (Hugging Face Hub, v 3.1).

After transcription and speaker annotation, we extracted sentence embeddings using an LLM and converted each utterance into a 768-dimensional vector using Sentence-BERT (multilingual-MiniLM L12 v2). A cap was set on the number of utterances per speaker and utterances were randomly sampled from those who exceeded the limit to avoid dominance by speakers who spoke excessively.

For the dimensionality reduction of high-dimensional vectors, we utilized UMAP, a widely recognized technique, and compared the results. We adopted HDBSCAN [13], a robust clustering method, to classify speaker characteristics after dimensionality reduction. HDBSCAN does not require a prior specification of the number of clusters and can automatically detect noise points, isolating outlier sentences.

These methods are commonly used in active learning for large-scale datasets [14]; UMAP and t-SNE are widely applied to visualize the distribution of long-tailed data, whereas K-means is a standard method for visualizing cluster variability. Such techniques are generally applicable and stable for LLM-based embedding vectors [15][16].

We then measured the divergence between each speaker's distribution and the overall topic distribution using the Jensen–Shannon distance [17], quantifying the extent to which each speaker deviated from the group topic distribution. By combining the minimum utterance threshold with the divergence in distance, we determined whether speakers exhibited outlier behavior in their utterances.

By setting a threshold in advance, only those with substantial deviations were identified as outliers, whereas the others were correctly recognized as normal, thus avoiding unnecessary separation of speakers with slightly different tendencies—a key improvement over previous studies. We applied these methods to both previously used and newly collected datasets with different speakers to verify their applicability.

Operationally, let d_s denote the Jensen–Shannon distance for speaker s and let μ and σ be the mean and (unbiased) standard deviation of $\{d_s\}$ across speakers in the same meeting. We flagged a speaker as "divergent" iff $d_s > \mu + K\sigma$ with $K=1.0$, and only if the speaker produced more than $N_0 = 10$ utterances.

In a sensitivity analysis, we also examined $K \in \{2.0,3.0\}$ and found that the qualitative conclusions remained unchanged; larger K yields a more conservative detector, while N_0 primarily controls robustness against low-activity speakers.

C. Analysis Results

Figs. 1 and 2 show examples of the datasets analyzed using the proposed method. We analyzed data from two separate meetings.

```
=== Speaker Stats ===
speaker2   JS=0.489  N=  14  H=0.99 <-- OUTLIER
speaker4   JS=0.225  N=  39  H=3.09
speaker1   JS=0.218  N=  36  H=3.35
speaker5   JS=0.214  N=  39  H=3.36
```

Fig. 1. Identification of a speaker with a markedly different tendency in the first meeting using the proposed method.

```
=== Speaker Stats ===
speaker2   JS=0.159  N=  40  H=0.85 <-- OUTLIER
speaker3   JS=0.125  N=   3  H=0.92
speaker1   JS=0.056  N=  42  H=0.99
speaker4   JS=0.025  N=  35  H=0.99
```

Fig. 2. Identification of a speaker with a markedly different tendency in the second meeting using the proposed method.

The results represent the final outputs obtained using the proposed method. However, by visualizing the intermediate output of UMAP in two dimensions during the computational process, one can observe specific differences in the tendencies among speakers. Fig. 3 shows the results of visualizing the UMAP output in two-dimensions. Four major clusters are observed in this visualization. Although all speakers except Speaker 2 had utterances distributed across multiple clusters, Speaker 2's utterances were concentrated within a particular cluster, indicating a biased distribution. Such a visualization is important for verifying the analysis process and examining the relationship between utterance tendencies and the vocabulary used.

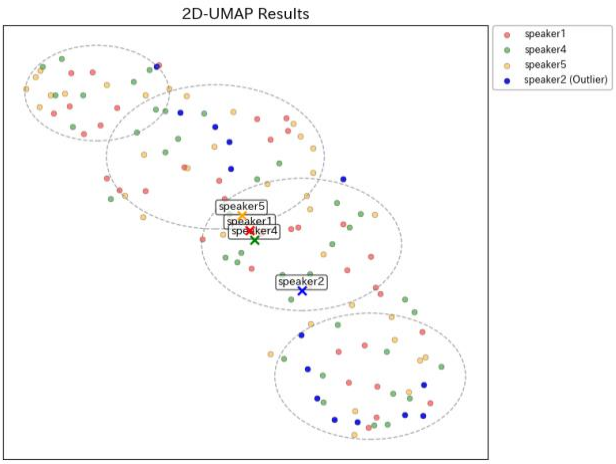


Fig. 3. Visualization of the clustering results in the 2-D UMAP projection

Furthermore, the proposed method requires only 21 s to execute, including the time required to download the necessary models.

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V. DISCUSSION

First, we discuss the theoretical implications. The proposed method enables a stable analysis that avoids unnecessary separation of speakers when no true outliers are present, by combining LLM-derived embedding vectors with UMAP and clustering techniques. This is an improvement over previous studies[5][6] in which concerns about the generalizability of PCA remained. However, this remains a methodological hypothesis, and future work should apply the framework to a larger and more diverse set of data to further examine its generalizability and potential limitations. This study is based on a single-company case study with a limited number of meetings, which constrains external validity. Conversational norms, language use, and organizational context in a seed-stage high-tech firm may not fully generalize to other settings. Our pipeline also relies on a single embedding model and a specific configuration of UMAP and HDBSCAN. Errors from ASR transcription and speaker diarization may propagate into the estimated topic distributions and the JS-based divergence. As future work, we plan to replicate the analysis across multiple companies and a larger set of meetings over longer time spans, compare alternative embedding models and clustering configurations and hyperparameters, and include human-in-the-loop assessments to verify whether flagged "divergent" speakers align with expert judgments.

In this study, OpenAI's Whisper, a well-known LLM, was used for speech transcription, and pyannote was employed for speaker diarization. This is because Whisper's speaker separation currently does not achieve sufficient accuracy for Japanese recordings, necessitating the use of different models for transcription and diarization. Consequently, end users must download and set up multiple models and environments. With the anticipated improvements in the API offerings and model performance for Whisper, such processes are expected to eventually be integrated into a single API, further enhancing usability.

Furthermore, although UMAP was utilized as the dimensionality reduction method in this study, various other nonlinear dimensionality reduction techniques exist. Determining the most generalizable and best suited method for this framework, particularly as more datasets are incorporated, remains an open issue and a subject for future research.

In this study, the threshold was manually determined based on the total number of utterances. However, as more data become available for validation in the future, it will be important to develop an automatic mechanism for determining the threshold, such as setting it as a fixed percentage of the total number of utterances.

Next, we discuss the practical implications. Previous studies [5][6] suggested a relationship between tolerance for uncertainty and a tendency to speak predominantly in technical terms. Our findings indicate that this relationship may result not only from a bias toward technical terminology but also from a marked paucity of utterances containing words associated with specific clusters—that is, a lack of discourse on certain topics. However, since this pattern may be a secondary effect of the predominance of technical terms, future research should develop and apply methods to determine whether this pattern is statistically significant.

Beyond the immediate methodological gain, this study also advances a theoretical account of how modern NLP can enhance organizational analytics. In particular, LLM-based sentence embeddings act as an intermediate representation that translates qualitative meeting content into stable, comparable signals at the organizational level. By projecting utterances onto a shared semantic manifold (UMAP) and summarizing each speaker as a probability distribution over data-driven topics, we can operationalize constructs such as topical breadth, specialization, and divergence. The Jensen–Shannon–based divergence then provides an interpretable 0–1 measure of distance from the meeting's normative discourse. Taken together, this pipeline bridges the long-standing gap between qualitative conversation analysis and quantitative HR decisions, for example by informing assessments of collaboration risk, information silos, and role alignment.

VI. CONCLUSION

In this study, we propose a framework that replaces PCA in previous research with a more robust approach that combines LLM-derived embedding vectors for stable word vectorization independent of frequency, with nonlinear dimensionality reduction and clustering techniques. Speakers with extremely few utterances are not forcibly identified as outliers solely based on utterance count, but are evaluated in terms of the content of their speech, unlike PCA-based methods. In addition, in experiments with existing datasets, the proposed method successfully identifies speakers exhibiting different behaviors, which is consistent with previous research.

A future challenge is to further investigate the applicability and generalizability of this framework by applying it to an expanded range of datasets.

Moreover, the use of UMAP enables us to observe variations in the distribution of utterances that cannot be captured using PCA. Specifically, speakers identified as outliers have conversations biased toward topics, whereas other speakers contribute utterances more evenly across all topic clusters. This suggests that the frequent use of technical terms and the absence of discussion on certain topics or topic bias may contribute to conversational misalignment within the organization, as well as to the tolerance for uncertainty described in previous studies.

In future, further validation with additional speakers and different meeting contexts is expected to lead to updates and a broader generalization of the proposed method.

Furthermore, while this study focused on generalization and application within a single company,

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the proposed methodology is readily transferable to other organizations without modification. Therefore, in future studies, we plan to validate its external validity using open datasets and data from other organizations.

ACKNOWLEDGMENT

The authors would like to express their gratitude to the management team of Adansons Inc. for granting permission to use the data anonymously. This study was supported by JPPS KAKENHI (grant numbers JP22K01742).

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